

***Ethiopian Giftedness and Talent Development Center Advanced
Electronics & Robotics Track: Final Project Documentation***



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Smart Assistive Cane for Visually Impaired Navigation

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Institution: Ethiopian Giftedness and Talent Development Center

Program: EGATE Training Program – Advanced Electronics & Robotics Track

1. Executive Summary & Project Objectives

This project presents an AI-powered smart assistive cane designed to enhance mobility, environmental awareness, and personal safety for visually impaired individuals. Traditional white canes only detect floor-level physical contact obstacles. This system overcomes that fundamental limitation by integrating **MiDaS monocular depth estimation**, **YOLOv8 real-time object detection**, an **analog IR proximity sensor**, and **multithreaded audio-haptic feedback** into a unified hardware-software platform.

- **Multithreaded Spatial Awareness:** Process continuous object detection (YOLOv8) and 3D terrain hazard analysis (MiDaS) asynchronously without camera feed stuttering or frame drops.
- **Strict Navigational Timing:** Provide turn-by-turn guidance prompts on a dedicated background thread clock every 10 seconds.
- **Hierarchical Audio & Haptic Feedback:** Prioritize real-time critical hazards (holes, overhead obstacles, vehicles) over timed turn-by-turn directions to prevent speech clutter.
- **Low-Latency Microcontroller Control:** Maintain non-blocking hardware control on an Arduino Mega 2560 using proportional PWM haptic motor feedback and dynamic IR distance sensing.

2. Computer Vision & Advanced Feature Specifications

- **MiDaS 3D Monocular Depth Sensing:** Replaces physical ultrasound with neural depth mapping. The incoming frame is divided into three vertical scan zones:
 - **Ground Zone:** Evaluates depth variance calculations (greater than 0.10) to flag sudden drop-offs, holes, and stairs.
 - **Center Zone:** Identifies large walls, doors, or blocking barriers (greater than 0.80 relative depth).
 - **Overhead Zone:** Alerts the user to low-hanging branches, beams, or signboards (greater than 0.75 relative depth).

- **Decoupled 10-Second Guidance Clock:** Navigation prompts execute on an isolated background thread timer. Turn-by-turn instructions issue strictly every 10 seconds without locking or delaying vision processing loops.
- **Non-Blocking Multithreaded Speech Engine:** Replaces blocking audio calls with a thread-safe pytsx3 queue. Immediate safety alerts preemptively terminate and replace timed navigation cues.
- **Hybrid Haptic-Audio Priority Matrix:** Combines high-priority spoken warnings with proportional handle vibrations generated by the Arduino Mega.

3. Multi-Tier Hazard Priority Matrix

Tier	Priority Level	Hazard / Trigger Source	Visual / Sensor Input	Spoken Audio Reaction	Haptic Motor Signal
Tier 0	0 (Critical)	Hole / Drop-Off / Barrier	MiDaS Depth Matrix	Spoken instantly ("Caution, drop off ahead")	Max Speed (255 PWM)
Tier 1	1 (High)	Vehicle / Moving Hazard	YOLOv8 Pipeline	Interrupts lower audio ("Car straight ahead")	Full Speed (255 PWM)
Tier 2	1 (Medium)	Physical Proximity Obstacle	Sharp IR Sensor (10 cm to 50 cm)	Position/Distance speech alert	Linear Mapping (120 to 255 PWM)
Tier 3	2 (Low)	Scheduled Turn Navigation	10-Second Guidance Clock	Triggers every 10s ("Walk straight 20 meters")	Motor Idle (0 PWM)

4. Hardware Architecture & Bill of Materials (BOM)

Item No.	Component Name	Specification / Model	Quantity	Role / Description
1	Microcontroller / MCU	Arduino Mega 2560 Rev3 (ATmega2560)	1	Sensor sampling & PWM motor output control
2	Distance Sensor Module	Sharp GP2Y0A21YK0F Analog IR (10 cm to 80 cm)	1	Ground proximity detection
3	Actuator / Motor	Mini DC Vibration / Haptic Motor (5V)	1	Handle haptic feedback
4	Transistor Driver	2N2222 / TIP120 NPN BJT Transistor	1	Low-side motor driver circuit
5	Diode Protection	1N4001 Flyback Protection Diode	1	Suppresses motor back-EMF voltage spikes
6	Resistor	1 kOhm, 1/4W Carbon Film	1	Base current limiting for NPN transistor

7	Power Supply	External 5V Power Module / Battery Pack & USB Line	1	Common power rail for MCU and motor driver
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5. Circuit Design & Electrical Wiring Safeguards

- **UART Communications:** Arduino Mega Serial 0 connects over USB (9600 Baud) to the host Python execution environment.
- **Analog Sensing:** Sharp IR sensor signal line connects to analog pin **A2**, powered via the 5V rail with a shared ground.
- **Haptic Driver Circuit:** Arduino PWM pin **9** connects through a 1 kOhm resistor to the NPN transistor base. The motor sits between the +5V supply rail and the transistor collector.
- **Inductive Protection & Voltage Spike Mitigation:** Rapid PWM motor switching initially induced back-EMF voltage spikes that caused transient voltage sags, freezing the microcontroller. A 1N4001 flyback diode was wired in reverse bias across the motor leads (cathode to +5V, anode to transistor collector) to suppress voltage spikes and maintain MCU stability.

6. Firmware Implementation & Control Logic Summary

- **Software Stack / IDE:** Arduino IDE (C++ embedded firmware), Python 3.11, OpenCV (cv2), PySerial, and Ultralytics YOLOv8.
- **Non-Blocking Logic Loop:** The firmware uses non-blocking millis() polling intervals instead of blocking delay() calls. The analog IR sensor on pin **A2** is sampled every 80 ms, averaged across 5 readings, and converted to distance in centimeters.
- **Serial Buffer Management & Watchdog:** The serial read logic flushes outdated bytes in the input buffer to retain only the newest command. A 1500 ms watchdog timer automatically resets the state if serial communication drops.

7. Testing, Performance Metrics & Conclusion

- **Less than 100 ms Response Latency:** Asynchronous thread queues and non-blocking serial buffer flushing reduced motor and speech reaction times to under 100 ms upon hazard detection.
- **10 cm to 50 cm Detection Bounds:** Analog IR range mapping filtered out floating pin noise (less than 90 raw ADC) and close-range optical distortion (greater than 650 raw ADC), achieving reliable proximity sensing.
- **60 ms Serial Throttling:** Command-change filtering prevented serial port buffer congestion during continuous vision inference loops.
- **Commercialization & Scaling Roadmap:** Future milestones focus on moving from breadboard wiring to a custom PCB layout, migrating the Python computer vision stack

to an onboard Raspberry Pi micro-PC, and producing a 3D-printed, weather-resistant outer enclosure.